

APPLETON, WI, USA

WMO: 726457

Lat: 44.267N

Lon: 88.517W

Elev: 917

StdP: 14.22

Time Zone: -6.00 (NAC)

Period: 92-19

WBAN: 04825

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.0	(c) -0.3	(d) -14.5	(e) 2.6	(f) -4.7	(g) -8.7	(h) 3.6	(i) 1.2	(j) 27.6	(k) 17.6	(l) 24.8	(m) 22.6	(n) 10.6	(o) 290	(p) 0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 17.6	(c) 88.3	(d) 75.1	(e) 84.5	(f) 72.3	(g) 82.2	(h) 70.6	(i) 77.6	(j) 85.2	(k) 75.3	(l) 82.4	(m) 73.5	(n) 79.7	(o) 11.9	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.1	136.3	81.4	73.0	126.6	79.4	71.9	121.8	78.2	41.5	85.7	39.5	82.1	37.7	79.0	91.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 24.7	(b) 21.6	(c) 19.0	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 24.7	(b) 21.6	(c) 19.0	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.9	19.1	24.7	32.7	46.4	56.3	66.7	71.9	69.4	62.7	49.7	37.0	24.9
	DBStd	20.20	10.80	11.20	10.40	8.81	8.28	7.54	5.53	5.70	7.69	8.87	9.37	11.71
	HDD50	3710	959	709	543	168	32	0	0	0	5	115	400	779
	HDD65	7210	1424	1129	1000	562	292	72	10	24	134	480	841	1242
	CDD50	2578	0	0	8	59	226	501	679	601	387	107	9	1
	CDD65	602	0	0	0	3	21	122	223	160	66	7	0	0
	CDH74	4507	0	0	1	43	200	980	1777	1027	430	49	0	0
	CDH80	1204	0	0	0	8	42	274	533	262	80	5	0	0
	WSAvg	9.6	10.7	10.3	9.9	11.2	10.3	8.6	8.0	7.4	8.9	9.7	10.2	10.0
	PrecAvg	31.9	1.3	1.0	1.8	2.8	3.7	4.7	3.9	3.5	3.1	2.5	1.8	1.5
(15) Precipitation	PrecMax	41.3	2.8	1.8	3.7	5.3	7.6	8.9	11.5	9.2	5.9	5.8	4.2	3.3
	PrecMin	27.9	0.3	0.3	0.2	0.6	1.6	1.5	1.1	1.7	0.6	0.3	0.2	0.2
	PrecStd	3.6	0.7	0.4	1.0	1.1	1.5	2.2	2.5	1.6	1.3	1.4	1.1	0.9
	PrecStd	3.6	0.7	0.4	1.0	1.1	1.5	2.2	2.5	1.6	1.3	1.4	1.1	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.9	52.5	66.4	79.5	84.4	90.7	93.3	90.6	86.0	80.7	65.7	56.8
		MCWB	40.4	50.7	55.6	61.1	68.1	73.4	80.2	77.8	72.4	68.8	56.4	54.2
	2%	DB	39.5	45.5	57.5	72.4	80.4	86.3	89.7	86.0	81.9	74.6	60.7	49.7
		MCWB	35.3	41.0	49.9	58.7	66.2	72.8	76.5	74.5	69.6	64.3	53.6	48.6
	5%	DB	36.2	41.9	53.7	65.8	74.9	82.8	85.7	82.1	79.1	70.0	55.1	43.3
		MCWB	34.5	38.2	47.9	55.4	62.7	71.2	74.0	71.6	67.2	62.4	48.7	39.8
	10%	DB	34.0	38.6	47.0	61.3	71.4	80.8	82.3	80.6	75.1	63.7	51.7	38.8
		MCWB	32.9	35.9	42.7	49.8	59.6	69.5	70.7	70.1	66.0	57.2	47.3	36.2
	0.4%	WB	40.8	50.5	58.6	65.8	72.1	78.3	83.8	79.9	74.7	70.8	58.3	54.5
		MCDB	46.3	53.4	63.1	75.4	80.8	86.2	91.4	86.8	81.3	77.2	63.4	56.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.4	41.7	53.4	60.9	68.0	75.4	78.5	76.3	72.4	66.9	54.8	47.5
		MCDB	39.1	44.9	55.8	69.6	76.7	83.4	85.3	83.3	78.8	73.1	59.0	48.9
	5%	WB	34.5	38.5	47.5	56.6	65.0	73.1	76.1	74.3	70.2	62.7	50.7	40.2
		MCDB	36.1	41.4	51.9	64.1	72.4	81.2	83.2	80.2	76.1	67.5	54.3	42.6
	10%	WB	32.4	36.3	42.8	52.1	62.3	70.9	73.9	72.0	68.0	58.5	46.4	36.7
		MCDB	33.7	38.6	47.9	59.8	68.3	78.0	79.9	77.5	74.0	64.4	50.0	39.1
(35) Mean Daily Temperature Range	5% DB	MDBR	13.3	13.4	15.6	18.9	17.9	18.0	17.6	16.6	18.6	16.9	13.8	12.9
		MCDBR	14.4	15.1	22.4	27.2	22.8	20.5	20.2	18.9	20.6	23.4	20.4	15.8
		MCWBR	12.4	11.8	15.3	16.1	12.7	10.7	10.0	10.2	10.5	13.8	14.8	12.9
	5% WB	MCDBR	13.4	14.5	22.1	24.0	19.7	19.0	18.2	17.0	18.1	18.6	18.2	14.6
		MCWBR	11.9	11.7	16.9	16.7	12.9	11.5	10.3	11.0	10.8	13.7	15.5	13.0
(40) Clear-Sky Solar Irradiance	taub		0.265	0.320	0.330	0.370	0.400	0.406	0.392	0.393	0.365	0.325	0.294	0.270
	taud		2.402	2.217	2.363	2.311	2.269	2.288	2.331	2.322	2.391	2.503	2.516	2.444
	Ebn at Noon		276	272	287	283	276	274	277	272	272	271	264	263
	Edh at Noon		23	34	34	39	42	41	39	38	33	26	21	21
	RadAvg		516	856	1222	1445	1737	1897	1946	1660	1324	838	536	385
(45) All-Sky Solar Radiation	RadStd		59	81	131	145	137	135	90	109	103	87	49	39

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page